Lab 12 Advanced SQL 2

United International College

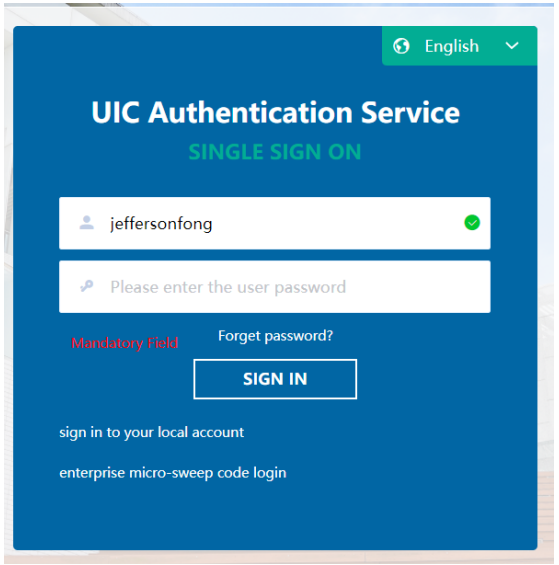
COMP3013 DBMS

Outline

- **Triggers**
  - Triggers - Event, Condition and Action
  - Triggers in SQL
  - Triggers Example, Row-Level and Statement-Level
  - Tricky Design Issues
  - Triggers Example, More
  - Triggers in MySQL, More
  - Triggers Example with UIC Database
- **Revisions**

Triggers – Event, Condition and Action

- For some simple condition checking within one table, we can use **`CHECK`**. But, for a complicated predicate crossing multiple tables, we need to use **`TRIGGER`**.
- A trigger is a statement that the system executes automatically as a side effect of a modification to the database.
- Reference: MySQL reference manual and [5.3] of textbook.
- We need two things:
  - Specify when a trigger is to be executed, event and condition.
  - Specify the actions to be taken when the trigger executes.
- Event-Condition-Action (non-database) example
  - Event – Click “SIGN IN” button
  - Condition1 – User name and password are correct
  - Action1 – Display the user’s UIC webpage
  - A different condition would trigger another action.



Triggers – Event, Condition and Action

Event-Condition-Action rules:

- When event occurs, check the condition; if true, do the actions.
- Move monitoring logic from apps into DBMS.
- Enforce constraints
  - beyond what constraint system supports and
  - automatic constraint “repair”.

Syntax

- An trigger in SQL takes the form

```
CREATE TRIGGER <trigger-name>
BEFORE/AFTER <event>
FOR EACH ROW
<Trigger body>
```

- **BEFORE** or **AFTER** specifies when the trigger is activated.
- <event> can be **INSERT**, **DELETE**, or **UPDATE**.
- **FOR EACH ROW** is a fixed phrase.
- <Trigger body> specifies the actions will be taken if some conditions are satisfied.

Syntax

The body of a trigger can contains

- referencing variables
  - **OLD.**<column name> refers to old rows for deletion or update events.
  - **NEW.**<column name> refers to new rows for insertion or update events.
- data modification statements
  - **INSERT**, **DELETE**, or **UPDATE** like other SQL statements
- Construct
  - Quoted by **BEGIN** and **END**
  - Details will be given later.

Example with UIC Database

- Referential Integrity

```
ALTER TABLE student ADD CONSTRAINT student_program
FOREIGN KEY p_code REFERENCES program(p_code)
ON DELETE CASECADE
```

- which can be implemented using a trigger

```
CREATE TRIGGER delete_program_student
AFTER DELETE ON program
FOR EACH ROW
DELETE FROM student
      WHERE student.p_code = old.p_code
```

Construct

- Some complicated constraints on multiple tables have to use **`BEGIN…END`** construct.
- Each statement in the construct must be ended with “;”
- Which creates a conflict to the delimiter of an SQL query.
- Thus, **`DELIMITER`** is used to change the delimiter before and after a trigger in SQL.
- For example, we change the delimiter from “;” to “|”.

```
DELIMITER |
<trigger> |
DELIMITER ;
```

Conditional Statement

- Suppose we want to make sure that every CST student has a GPA at least 1.0
- A conditional statement can be used.
- The entire trigger is

```
DELIMITER |
CREATE TRIGGER cst_student
AFTER INSERT ON student
FOR EACH ROW
BEGIN
      IF new.gpa<1.0 AND new.p_code IN (
            SELECT p_code FROM program WHERE p_name = 'Computer Science'
      ) THEN
            DELETE FROM student WHERE student.id = new.id;
      END IF;
END;|
DELIMITER ;
```

Exercises

- Create trigger(s) to make sure each person has to be either an instructor or a student but not both. (Total and disjoint generalization in the ER diagram.)
- Also create trigger(s) to make sure each person in *borrow* and *contact* already exists in either *instructor* or *student* but not both.
- A student gains the credits from a course if he/she receives passing grade (not 'F') from the course. A student graduates from the college if he/she obtains 130 credits. Create trigger(s) to on *enroll* to detect the graduates and remove those graduates from the database.

Save your queries in a txt file. Rename it as "COMP3013 Lab12 ###.txt", where "###" is your student ID. And submit it on iSpace. The DDL is 24 hours after the lab.

End of Lab 11